DE–3938 **11**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 =100)

1. (a) Discuss about the Ethics of Information Technology. (10)

   (b) Explain the Six elements of a computer and communication systems in detail. (10)

2. (a) Describe the connectivity and interactivity in computer and communication technology in detail. (10)

   (b) Explain the development in computer technology in detail. (10)

3. (a) Describe the four types of application software in detail. (10)

   (b) Discuss about the word processing and spreadsheets. (10)

4. (a) Write in detail about database software and presentation graphics software. (10)

   (b) Explain the communication software and Internet web browsers in detail. (10)

5. (a) Discuss about video confering and picture phones. (10)

(b) Write in detail about analog signals and digital signals with diagrams. (10)

6. (a) Explain the various factors affecting data transmission in datail. (10)

(b) Discuss about ISDN lines and cable modems. (10)

7. (a) Explain the hard disks and optical disks in detail with diagram. (10)

(b) Describe the types of database organization in detail. (10)

8. (a) Write in detail about the six phases of system analysis and design. (10)

(b) Discuss about the five generations of programming languages. (10)

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the decimal number system and hexadecimal number systems in detail. (10)

   (b) Discuss about number codes and character codes. (10)

2. (a) Simplify the boolean function F=A'B'C'+B'CD'+A'BCD'+AB'C' using four variable karnaugh map. (10)

   (b) Describe the 4-to-1 line multiplexer in detail with its function table. (10)

3. (a) Explain the parallel binary adder in detail with its function table. (10)

   (b) Describe the construction of arithmetic logic unit in detail. (10)

4. (a) Discuss about positive and negative numbers with examples. (10)

(b) Write in detail about the addition and subtraction in a parallel arithmetic element. (10)

5. (a) Explain the binary counter in detail with its function table. (10)

(b) Describe the linear select memory organization in detail. (10)

6. (a) Write in detail about the magnetic tape memories and magnetic disk memories. (10)

(b) Discuss about bidirectional shift register with parallel load with diagram. (10)

7. (a) Explain the controlling arithmetic operations in detail. (10)

(b) Discuss about I/O interface and I/O processors. (10)

8. (a) Discuss about programmed I/O and instruction word. (10)

(b) Write in detail about interrupts and interrupt priorities. (10)

DE–3940 **13**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. (5 × 20 =100)

All questions carry equal marks.

1. (a) Discuss about character set, identifiers and keywords in C with examples. (10)

   (b) Explain the data types, constants and variables in C in detail with examples. (10)

2. (a) Write in detail about 'break', 'continue' and 'go to' statements in C with example programs. (10)

   (b) Describe the 'do while' and 'for loop' statement in detail in C with example programs. (10)

3. (a) Explain the passing arguments to a function in detail in C with example programs. (10)

   (b) Describe the any four string library functions available in C in detail with example programs. (10)

4. (a) Write in detail about multidimensional array in C using matrix multiplication as an example program. (10)

(b) Discuss about the concept of defining and processing an array in C with example programs. (10)

5. (a) Explain the passing pointers to a function in detail in C with example programs. (10)

(b) Write in detail about the concept of dynamic memory allocation in C with example programs. (10)

6. (a) Describe the concept of structure in detail in C with example programs. (10)

(b) Discuss about the array of pointers in C with example programs. (10)

7. (a) Explain the stack structure in detail with its operations. (10)

(b) Describe the Queue structure in detail with its operations. (10)

8. (a) Discuss about singly linked list with its operations. (10)

(b) Write in detail about representing list as binary trees. (10)

DE–3941 **14**

DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION, MAY 2012.

First Semester

OFFICE AUTOMATION

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about various file operations in MS-Office.

   (b) Write notes on toolbars and dialog boxes in MS-Office.

2. (a) Write notes on the following

   (i) Moving and copying Text

   (ii) Find and Replace Text.

   (b) Explain about printing options in MS-Word.

3. (a) Write about bullets and numbering tool in MS-Word.

   (b) Discuss about table manipulation in MS-Word.

4. (a) Explain about number formatting in MS-Excel.

   (b) Discuss about different types of functions available in MS-Excel.

5. (a) Write notes on types of chart in MS-Excel.

   (b) Explain about Excel Databases.

6. (a) Discuss about PowerPoint toolbars.

   (b) Explain about formatting text in MS-PowerPoint.

7. (a) How will you add speaker notes in PowerPoint? Explain.

   (b) Discuss about Importing and Exporting Data in MS-Access.

8. (a) Explain about Field Property Settings.

   (b) Discuss on Embedding capability of MS-Office.

DE–3942 **21**

DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION, MAY 2012.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) (i) Discuss the features of object-oriented programming. (5)

(ii) What are the benefits of object oriented programming? (5)

(b) (i) Describe the major parts of a C++ program. (5)

(ii) Enumerate the rules of naming variables in C++. (5)

2. (a) (i) Discuss the data types in C++ with examples. (6)

(ii) What is a variable? With an example discuss on dynamic initialization of a variable. (4)

(b) List the operators in C++. Discuss with examples. (10)

3. (a) What is an expression? List the different types of expressions. Discuss any three with examples. (10)

(b) With suitable examples discuss any two control structures of C++. (10)

4. (a) How do structures differ in C and C++? Discuss. (8)

(b) Write a C++ program to read ten numbers through the keyboard and arrange the numbers in the increasing order. (12)

5. (a) What is function? What is the main advantage of passing arguments by reference. (10)

(b) What do you mean by overloading of a function? Discuss with example. (10)

6. (a) (i) Write a macro that obtains the largest of three numbers. (5)

(ii) What is modularization? Discuss. (5)

(b) (i) What are objects? How are they created? (4)

(ii) What is friend function? What are the merits and demerits of using friend functions? (6)

7. (a) (i) What is constructor? List the properties of the constructor functions. (5)

(ii) With an example discuss overloading binary operators using friend function. (5)

(b) What do you mean by inheritance in C++? What are the different forms of inheritance? Discuss with examples. (10)

8. (a) (i) Describe the various classes available for file operations. (5)

(ii) Describe the various approaches by which we can detect the end-of-file condition successfully. (5)

(b) (i) List the error handling functions. Discuss any two. (5)

(ii) Write a short note on command-line arguments. (5)

DE–3943 **22**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Calender Year Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. ($5 \times 20 = 100$)

1. (a) Explain the characteristics of a system in detail with example diagrams. (10)

   (b) Describe the computer–based Information systems in detail with example diagrams. (10)

2. (a) Write in detail about the Impetus for system change in system development cycle with diagram. (10)

   (b) Describe about the multifaceted role of the analyst. (10)

3. (a) Describe any one example case scenario in system analysis in detail. (10)

   (b) Discuss about interview, questionnaire and Record review. (10)

4. (a) Write in detail about Fact Analysis with example diagrams. (10)

(b) Discuss about Decision table and Decision tree with example diagrams. (10)

5. (a) Explain the logical and physical design in detail. (10)

(b) Discuss about the requirements of Forms design and Form Control. (10)

6. (a) Write in detail about major development activities with diagrams. (10)

(b) Describe the capturing data for input and Input Validation in detail. (10)

7. (a) Explain the program flow chart and system flow chart in detail with example. (10)

(b) Describe the system testing and Quality Assurance in detail. (10)

8. (a) Write in detail about the major phases in hardware and software selection. (10)

(b) Discuss in detail about post–implementation review with example diagrams. (10)

DE–3944 **23**

DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION, MAY 2012.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All question carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the overview of how to develop a visual basic application. (10)

   (b) Write in detail about the various Toolbars and its purposes. (10)

2. (a) Describe the Toolbox and Custom components in detail. (10)

   (b) Explain the Common Form Properties in detail. (10)

3. (a) Describe the Text boxes and Labels in detail. (10)

   (b) Discuss about creating controls in Visual Basic. (10)

4. (a) Explain the various data types available in Visual Basic in detail. (10)

   (b) Write in detail about displaying information on a form. (10)

5. (a) Discuss about the Numeric Functions and Financial Functions in Visual Basic. (10)

(b) Describe the Sub Procedure in Visual Basic in detail. (10)

6. (a) Explain the Control Arrays in detail in Visual Basic. (10)

(b) Describe the Flex Grid control in detail. (10)

7. (a) Write in detail about the object browser. (10)

(b) Discuss about Screen Scales and Animation in Visual Basic graphics. (10)

8. (a) Explain the file system controls in detail. (10)

(b) Describe the building ActiveX controls in Visual Basic in detail. (10)

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2012.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) Explain any three Input devices in detail with diagrams. (10)

   (b) Describe the various Hard Copy output devices in detail. (10)

2. (a) Describe the Bresenham's Line drawing Algorithm in detail with diagram. (10)

   (b) Discuss about drawing Ellipse and Circle Generation Algorithm. (10)

3. (a) Describe the two dimensional transformations (Reflection, Shear, Translation) in detail with diagrams. (10)

   (b) Write in detail about Windows and Viewport with diagrams. (10)

4. (a) Discuss about the point clipping and line segment clipping with diagrams. (10)

(b) Explain the Sutherland Hodgman Algorithm in detail with diagram. (10)

5. (a) Describe the three dimensional transformations (Scaling and Rotation) in detail with diagrams. (10)

(b) Write in detail about the composite transformations with diagram. (10)

6. (a) Explain the Cohen Sutherland Algorithm in detail with diagram. (10)

(b) Describe the Z-Buffer Algorithm in detail. (10)

7. (a) Write in detail about the specifying projection plane and view volumes. (10)

(b) Discuss about the concept of viewing process. (10)

8. (a) Explain the components of User Interface in detail. (10)

(b) Describe the styles of Command Language in detail. (10)

DE-4387 **15**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2012.

First Semester

LAB I — DATA STRUCTURES USING C

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a C program to find the roots of a quadratic equation using function definition $ax^2 + bx + c = 0$.

   (b) Write a C program to PRINT the grade of the students according to the following norms using switch statement

   Aggregate % of marks = 60 to 74.99 then grade is first class

   Aggregate % of marks = 75 to 100 then grade is first class with distinction

   Aggregate % of marks = 50 to 59.99 then grade is second class

   Aggregate % of marks = 1 to 49.99 then grade is fail

2. (a) Write a C program to PRINT the grade of the students according to the following norms using IF…THEN…ELSE statement

Aggregate % of marks = 60 to 74.99 then grade is first class

Aggregate % of marks = 75 to 100 then grade is first class with distinction

Aggregate % of marks = 50 to 59.99 then grade is second class

Aggregate % of marks = 1 to 49.99 then grade is fail.

(b) Write a C program to find the word is PALINDRAOME or not.

---------- Cut here ----------

3. (a) Write a C program to find the matrix addition for n×n matrix.

(b) Write a C program to do the following operations on stack (i) push (ii) pop.

---------- Cut here ----------

4. (a) Write a C program to sort a set of elements using insertion sort in ascending order.

(b) Write a C program to sort a set of elements using merge sort in ascending order.

---------- Cut here ----------

5. (a) Write a C program to read 10 values to an array variable. Use pointers to locate and display each value.

(b) Write a C program to find the factorial of a given number using recursion.

6. (a) A man is paid at the hourly rate of Rs. 15 per hour for the first 45 hours worked. there after, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for further hours worked. Write a C program to input the number of hours worked per week, calculate and print his gross weekly wage.

(b) Write a C program to implement insert and delete operations on queue using array concept.

---------- Cut here ----------

7. (a) Write a C program to copy contents of one file to another file.

(b) Write a C program to find the elements using linear search.

---------- Cut here ----------

8. (a) Write a C program to find the elements using binary search.

(b) Write a C program to convert a given number into words for number 1 to 5. Example 1 to ONE, 2 to TWO and 5 to FIVE.

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9. (a) Write a C program to implement the towers of Hanoi problem using recursion.

(b) Write a C program to sort the elements using bubble sort in ascending order.

10. (a) Write a C program to sort the elements using bubble sort in descending order.

(b) Write a C program to do the following operations using string built-in-function.

(i) String copy (copy one string to another string)

(ii) Find the length of the string

(iii) Compare the string.

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2012.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Answer any TWO questions One from SECTION A and another from SECTION B.

Each questions carries 50 marks.

SECTION A

1. Prepare an invitation for a function to be conducted in your institution using MS-WORD. Use different text orientation and pictures to make it attractive.

---------- Cut here ----------

2. Using MS-WORD type lecture notes and provide audio explanation with the help of sound files.

---------- Cut here ----------

3. Create a form letter using MS-WORD that informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal. Mailmerge it with an access database containing all the customer data. Prepare letters for customers whose due date falls in a specific range of dates.

4. Using MS-WORD prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

---------- Cut here ----------

5. Draw the bar chart and pie chart using MS-EXCEL for the data given below :

XYZ Company Ltd.

Year	Sales	Cost	Profit
1991	1000	400	600

Add data for 10 years from 1991 to 2000. Provide titles, legends, grids and data labels.

---------- Cut here ----------

6. Create two worksheets containing day to day house hold expenses for the months January and February 2011 with the following columns :

Item	Amount
Stationery	25.00

(a) Add 10 more different items as shown above, for two months in two different worksheets.
(b) Consolidate both the months data and find the total expenses on each item.
(c) Find the total expenses for two months.
(d) Find the maximum and minimum expenses amount.

---------- Cut here ----------

7. Create a worksheet with the following columns.
Salesman Number, Salesman Name, City, Product Name, Sale Amount.
Add three records from 5 different salesman who have carried out sales of different products in different cities. Find the following :
(a) Citywise Total Sales.
(b) Salesmanwise Total Sales.
(c) Productwise Total Sales.
(d) Grand Total Sales.

8. Create STUDENT table with the following fields REGNO, NAME, MARK1, MARK2, MARK3. Create ADDRESS table with fields REGNO, STREET, CITY and PIN.

   (a) Write a query to display REGNO, NAME and total of all the three subject marks.

   (b) Write a query to display REGNO, NAME, STREET, CITY, PIN and total of all the three subject marks.

---------- Cut here ----------

SECTION B

9. Create HTML page that divides the screen space into three frames (one vertical and two horizontal of equal size). Display three different files in them with scrolling titles relevant to their contents.

---------- Cut here ----------

10. Create HTML page to display the pictures of Elephant and Monkey. Link to another page when user clicks over the picture. The linked page should display features of the animal clicked.

---------- Cut here ----------

11. Create HTML page to display 10 courses and their highlighted features, conducted by an educational institution in a colourful format.

DE–4389 **25**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2012.

Second Semester

GRAPHICS USING C++ — LAB III

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Break -up of marks
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

The examiner should Select TWO questions and give it to the students by lot system.

1. Write a C++ program to draw a straight line using Bresenham's line drawing algorithm.

———————————————— cut here ————————————————

2. Write a C++ program to draw non linear object ellipse.

———————————————— cut here ————————————————

3. Write a C++ program to draw non-linear object circle and three concentric circle.

4. Develop a program to do the following :

   (a) Draw a triangle ABC, translate it to a distance and then rotate it through an angle clockwise.

   (b) With the same triangle reverse the operations as in case (a).

---------- cut here ----------

5. Write a C++ program to draw a straight line using DDA techniques.

---------- cut here ----------

6. Draw a ball. Perform animation on it.

---------- cut here ----------

7. Implement Bresenham's algorithm for drawing ellipse using C++ in graphics mode.

---------- cut here ----------

8. Write a C++ program to show graphical transformation on three dimensional object.

---------- cut here ----------

9. Write a C++ program to manipulate the graphical object using shading technique.

---------- cut here ----------

10. Implement the polygon clipping algorithm using C++ graphics functions.

DE–4390 **26**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2012.

Second Semester

LAB IV— VISUAL BASIC

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a VB project that receives a year number from a text box and month name from list box and displays number of days in the given month.

   Take care of leap years. Use lost focus event for list box.

   (b) Write a VB project that stores co employee records with fields EMPNO, NAME, AGE, SEX and SALARY, in an array. Display data fields in text boxes and provide command buttons to move to desired records.

2. (a) Write a VB project that receives a foreign currency values selected from a list box and converts it into equivalent Indian rupees. (Example USD 42.45, Sterling 71.30, D.Mark 25.52, SW Franc 31.58, SandiRiyal 11.40, French Franc 7.60, UAE Dhiram 11.55, Kuwait Dhinar 140.56)

(b) Write a VB project using control array that creates a scientific calculator with appropriate command buttons. Include the following capabilities for the calculator: +, –, *, /, %, power, square root, square and log (base 10).

---------- Cut here ----------

3. (a) Write a VB project to create a screen saver that displays a list of pictures with 1 second pause in between successive pictures.

(b) Write a VB project for commercial bank operations using SB account database with the following features.

(i) DEPOSIT AMOUNT

(ii) WITHDRAW AMOUNT(With minimum balance condition)

---------- Cut here ----------

4. (a) Write a VB project using built in AX control (Rich Text Box), develop the windows NOTEPAD like editor with File and Edit menus and also display the floating menu whenever necessary.

(b) Write a VB project for a blood bank that maintains a list of donors with address and their blood group. Provide the following reporting features:

Search and display the address of a particular donar, given the name in a text box.

5. (a) Write a VB project using AX DLL or EXE add a class module that would perform the following functions:

   (i) Test whether the given number is perfect or not.

   (ii) Find the factorial of the given number.

   (b) Write a VB project using ActiveX control to create a Textbox that accepts only numeric value. Provide the following properties for the text box: Back color, Fore color, and Text.

---------- Cut here ----------

6. (a) Create a VB application to make the text bold, italic underline and also to color the text (Using check box and option button)

   (b) Write a VB project to display a message (Example left/right mouse button is clicked) When the mouse button is clicked using mouse move event of the form.

---------- Cut here ----------

7. (a) Write a VB project to display the availability of seats when the user selects the course name from the Combo box and timings from another Combo box.

   (b) Design and implement one ActiveX control for digital clock.

8. (a) Write a VB project using AX DLL or EXE add a class module that would perform the following functions.

(i) Sum of digits

(ii) Whether the given number is Amstrong or not.

(b) Create an application using VB to move elements from one list to another list and add new items.

---------- Cut here ----------

9. (a) Write a VB project for a blood bank that maintains a list of donors with address and their blood group. Provide the following reporting features.

(i) Display all the donors (Using data reports)

(ii) in age group 20-30

(iii) in a particular city.

(b) Write a VB project to calculate the amount when the user selects the class 'type of the train' from the Combo box and also enters the number of passengers.

---------- Cut here ----------

10. (a) Write a VB project for a blood bank that maintains a list of donors with address and their blood group. Provide the following reporting features.

(i) Display all the donors (using data reports)

(ii) With particular blood group

(iii) Male donors with particular blood group

(iv) Female donors with particular blood group.

(b) Write a VB project for commercial bank operations using SB account database with the following features.

(i) ADD NEW ACCOUNT

(ii) Calculate simple interest and update balance taking average of last 6 month balance in the account

(iii) CLOSE ACCOUNT.

DE–3946 **31**

DISTANCE EDUCATION

M.C.A. (Semester)/M.C.A. (S)/(Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

DISCRETE MATHEMATICS

2005 to 2010 Calendar Year

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the concept of predicates with examples. (10)

   (b) Consider the statement "Given any positive integer, there is a greater positive integer" symbolize this statement with and without using the set of positive integers as the universe of discourse. (10)

2. (a) Obtain a conjunctive normal form of each of the formulas

   (i) $P \wedge (P \rightarrow Q)$

   (ii) $\neg(P \vee Q) \rightleftarrows (P \wedge Q)$. (10)

   (b) Obtain the principal disjunctive normal forms of

   (i) $\neg P \vee Q$

   (ii) $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$. (10)

3. (a) Show that $(R \subseteq S) \wedge (S \subset Q) \Rightarrow R \subset Q$ is it correct to replace $R \subset Q$ by $R \subseteq Q$? Explain your answer. (10)

(b) Discuss about the concept of power set with examples. (10)

4. (a) Explain the properties of relations using graphs of relations in detail with examples. (10)

(b) Let $X = \{1, 2, \ldots 7\}$ and

$$R = \{< X, Y > / x, y \text{ is divisible by } 3\}$$

Show that R is an equivalence relation. Draw the graph of R. (10)

5. (a) Define :

(i) Right-invertible and left invertible.

(ii) Idempotent

(iii) n-ary operation. (10)

(b) Define the term characteristic function of a set and show that $A \cap (B \cap C) = (A \cap B) \cup (A \cap C)$. (10)

6. (a) Discuss about the properties of Algebraic systems with examples. (10)

(b) Let $< G, * >$ and $< H, \Delta >$ be groups and $g : G \to H$ be a homomorphism. Then prove that the kernel of g is a normal subgroup. (10)

7. (a) Discuss about the concept of binary tree representation, positional m-ary tree and binary tree representation of forest with suitable examples. (10)

(b) (i) Prove that in a simple digraph, the length of any elementary path is less than or equal to $n-1$, where n is the number of nodes in the graph. Similarly, the length of any elementary cycle does not exceed n. (6)

(ii) Define the term path, simple path, elementary path and cycle. (4)

8. (a) Define the term group and give examples. (4)

(b) Prove that every finite group of order n isomorphic to a permutation group of degree n. (8)

(c) State and prove Lagrange's theorem. (8)

DE–3947 **32**

DISTANCE EDUCATION

M.C.A./M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5×20 = 100)

1. (a) (i) Discuss the history of the operating systems. (5)

(ii) Discuss the type of services provided by an operating systems. (5)

(b) (i) Compare the time-sharing and real time operating systems. (4)

(ii) Write a short note on operating system structure. (6)

2. (a) What are the issues related to interprocess communication? Discuss. (10)

(b) What is a semaphore? Discuss any two applications of semaphore. (10)

3. (a) What is priority scheduling? Compare any two process scheduling algorithms. (10)

(b) (i) What is message passing? Discuss. (5)

(ii) Write a short note on critical sections. (5)

4. (a) (i) What is a device driver? Discuss. (5)

(ii) What is the difference between deadlock and starvarion ? How can the system detect that some of the processes are straving. (5)

(b) (i) How deadlock can be prevented? Discuss. (5)

(ii) Write a short note on resource allocation graph. (5)

5. (a) (i) What is swapping? Discuss memory management without swapping or paging. (5)

(ii) What is page fault? How you can avoid page faults. (5)

(b) Discuss on the following

(i) Multiprogramming with variable partitions.

(ii) Memory management with bit maps. (5+5)

6. (a) What is a virtual memory? Discuss the virtual memory concept with necessary diagrams. (10)

(b) List the different page replacement algorithms. Compare any three. (10)

7. (a) What is a file? Describe the different operations on files. (8)

(b) Write a short note on different directory structures. (12)

8. (a) What are the goals of protection? Write a short note on different protection mechanism and policies. (10)

(b) Discuss on the following :

(i) File servers

(ii) Disk space management. (5 + 5)

DE–3948 **33**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (S)/ (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

RDBMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the different user interfaces in detail. (10)

   (b) Discuss about the features and advantages of DBMS. (10)

2. (a) Write in detail about schema and subschema with examples. (10)

   (b) Describe the database management system in detail with diagram. (10)

3. (a) Discuss about the concepts of relational model. (10)

   (b) Explain the data control facilities in relational model in detail. (10)

4. (a) Write in detail about data manipulation facilities in relational model. (10)

(b) Describe the data definition facilities in relational model in detail. (10)

5. (a) Explain the client/server systems concepts in Oracle in detail. (10)

(b) Describe the types of databases in detail. (10)

6. (a) Write in detail about oracle database architecture. (10)

(b) Discuss about the overview of oracle architecture. (10)

7. (a) Write in detail about DML commands with examples. (10)

(b) Describe the Joining multiple tables in a Query in Oracle in detail with examples. (10)

8. (a) Explain the Embedded SQL statements in detail with examples. (10)

(b) Discuss about renaming a table and dropping a table with examples. (10)

DE–3949 **34**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S)/(Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain about directory permission and file permission in UNIX.

   (b) Explain the following commands

      (i) pwd

      (ii) rmdir

      (iii) ls

      (iv) find

      (v) more.

2. (a) Discuss about filters in UNIX.

   (b) Explain the meaning of each of the following fields

      (i) Login name

      (ii) Password

      (iii) User id

(iv) Group id

(v) User information

(vi) Login shell

(vii) Home directory.

3. (a) Write notes on program output as an argument.

(b) Discuss in detail about variables in shell programming.

4. (a) Explain about I/O redirection.

(b) Discuss about while and until loops.

5. (a) Write about interrupt handling mechanisms in shell programming.

(b) Explain the following

(i) Overwrite

(ii) Zap

(iii) News

(iv) Pick commands.

6. (a) Discuss in detail about signal and interrupts in UNIX.

(b) Explain about accessing the environment in UNIX.

7. (a) Discuss about control flow and relational operators.

(b) Write notes on MS Macro Package.

8. (a) Explain about tbl and eqn preprocessors.

(b) Describe about document preparation.

DE–3950 **35**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the concepts of reading and various reading styles and its importance in developing communication skills in detail. (10)

   (b) Describe the Effective writing, Report writing and Agenda writing in detail with examples. (10)

2. (a) Write in detail about Speech writing, Letter writing and Article writing with examples. (10)

   (b) Discuss about the concept and definition of communication. Also explain its types in detail. (10)

3. (a) Write in detail about definition, importance and advantages of Listening. (10)

   (b) Describe the various types of Interviews in detail. (10)

4. (a) Explain the purpose and process of group discussion in detail. (10)

   (b) Discuss about the origin and development of body language. Also explain its types in detail. (10)

5. (a) Explain the meaning, importance and fundamentals of negotiation techniques in detail. (10)

(b) Describe the process and techniques for negotiation in detail. (10)

6. (a) Write in detail about meaning, importance and objectives of meetings. (10)

(b) Discuss about Mock meetings and seminars. (10)

7. (a) Describe the concept of Downward and Upward Communication in detail. (10)

(b) Explain the meaning and types of presentation in detail. (10)

8. (a) Discuss about the various techniques for delivering presentation and model presentation. (10)

(b) Discuss about the personal style and a postscript in presentation. (10)

DE–3951 **41**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) Explain the OSI Reference model of the network in detail with diagrams. (10)

   (b) Describe the Telephone system in detail with diagram. (10)

2. (a) Write in detail about the Architecture of Integrated service Digital Network with diagram. (10)

   (b) Discuss about the Asynchronous Transfer Mode. (10)

3. (a) Explain the Error detection and correction codes in detail. (10)

   (b) Describe the sliding window protocols in detail with diagrams. (10)

4. (a) Discuss about the concept of Petri Net Models. (10)

   (b) Write in detail about the SLIP and Point-to-Point Protocols with diagrams. (10)

5. (a) Explain in detail about any three network routing algorithms with diagrams. (10)

(b) Describe in detail about congestion control algorithms with diagrams. (10)

6. (a) Write in detail about the Fire Walls and its types with diagrams. (10)

(b) Discuss about the ATM LANS with diagram. (10)

7. (a) Explain the Transmission Control Protocol in detail. (10)

(b) Discuss about Multiplexing and crash recovery in transport layer with diagram. (10)

8. (a) Explain the Data Encryption Standard algorithm in detail with diagram. (10)

(b) Discuss about the Electronic Mail privacy. (10)

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DE–3952 **42**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Senthil limited has purchased and issued materials as under:

January 1	stock of materials : 200 units at Rs 2.50 per unit
January 3	purchased 300 units at Rs. 4 per unit
January 7	purchased 500 units at Rs.4 per unit
January 10	issued 600 units
January 12	purchased 400 units at Rs 4 per unit
January 18	issued 500 units
January 24	purchased 400 units at Rs 5 per unit
January 28	issued 200 units

Prepare stores ledger under FIFO methods.

2. From the following trial balance, prepare a balance sheet and revenue account of happy- go- lucky life insurance corporation for the year ended 31-12-1978.

	Dr	Cr
	Amount	Amount
Loans on Reversion	80,000	
Loans on life policies	6,40,000	
Municipal securities	4,00,000	
Government of India securities	6,40,000	
Foreign government securities	1,60,000	
Freehold	6,40,000	
Reversions	4,00,000	
Outstanding premiums	40,000	
Outstanding Interests,	10,000	
dividends etc		
Surrenders	14,800	
Cash in hand	40,000	
Claims admitted bur or not paid		8,000
Claims on business in India	1,60,000	
Claims outside India	8,000	

	Dr Amount	Cr Amount
Sundry creditors		12,000
Balance of life assurance fund on 1-1-1978		32,00,000
Expenses of management	16,000	
Premiums of business in India		3,20,000
Outside India		68,800
Mortgages on property in India	2,40,000	
Outside India	1,20,000	
	36,08,800	36,08,800

3. The following data are obtained from the records of prem ltd.

Years	2007 (Rs)	2008 (Rs)
Sales	80,000	90,000
Profit	10,000	14,000

Calculate p/v ratio and breakeven point in rupees.

4. From the following trial balance of the Marine Insurance Company Ltd. Prepare final accounts for the year ended December31, 1963.

Dr. Balances	Rs.
Claims paid : Marine	3,53,000
Claims paid and outstanding	
Fire	71,000

Accident and general: commission	2,66,000
Accident and general:	1,66,000
Fire	37,000
Expenses	Rs.
Fire	86,000
Accident and general	1,83,000
Marine	1,02,000
General	4,04,000
Preliminary expenses	4,10,000
Investments	20,50,000
Freehold and lease hold	
Premises	9,62,000
Agent's balances	12,83,000
Sundry debtors	2,70,000
Office furniture. Fittings	75,000
Interest accrued but not	
Payable	14,000
Cash in hand and at bank	7,50,000
Rs.	74,82,000
Cr. balances	Rs.
Share capital (6,90,000 shares of Rs.10 each Rs 5 per Share paid up)	34,69,000
Premium received:	
Fire, loss re-insurance	3,52,000
Marine, less commission and reinsurance	12,27,000

Cash received on forfeited shares	1,000
Miscellaneous receipts	5,000
Interests and dividends	
Less tax	1,05,000
Sundry creditors	1,89,000
Claims admitted but not paid:	
Fire	17,000
Accident and general	14,000
Due to reinsurances:	
Fire	12,57,000
Accident and general	8,41,000
Marine	5,000
Rs	74,82,000

The following adjustments are to be made before the preparation of the final accounts.

Freehold and leasehold properties are revalued on December 31, 1963.

And an increase of Rs. 1,98,000 is shown over cost of Rs. 9,62,000. This increase is to be credited to capital reserve, write off preliminary expenses Rs 1,10,000 and depreciation of investment Rs.39,000 against capital reserve, The balance of marine account is carried forward. Upon both fire account and accident and general account provision for unexpired risks is made at 35 percent of premiums received less reinsurance and the balance of these accounts carried to profit and loss account.

5. From the data pertaining to vasuprada limited given below, calculate the primary earnings per share and also the diluted earnings per share:

(a)	10,000 equity shares of Rs 100 each	Rs 10,00,000
(b)	10,000 10% preference shares of Rs 100 each	10,00,000
(c)	The preference shareholders have the right to convert 50% of their holding into equity shares	
(d)	Net income for the year under review	3,00,000

6. Srivatsasa limited was incorporated on 1st December, 1979. It is proposed to commence trading business on 1st January 1980 and the following are the budgeted details for the six months upto 30th June 1980.

(a) On 1st January the company will acquire buildings at a cost of Rs. 50,000, furniture at a cost of Rs.5000 and machinery at a cost of 25,000.

(b) Sales will be Rs.60,000 each month. The markup is 50% on cost.

(c) Overhead expenses will amount to Rs 10,000 each month except in April when they will amount to Rs. 15,000. All expenses are to be paid in the same month in which they are incurred.

(d) An initial stock of goods is to be bought on 1 st January at a cost of Rs. 15,000 and the same stock is to be maintained throughout.

(e) Trade creditors will be paid on the last day of the month in which the goods are bought Payments for all sales will be received on the last day of the month after the one in which the goods are sold.

(f) You are required

(i) To estimate the minimum capital required by the new company for the first 6 months

(Assume that there will be no borrowings during the period)

(ii) To prepare an income statement for the 6 months ending on 30th June 1980 after providing for depreciation on machinery and furniture at 10% per annum and .

(iii) To prepare the balance sheet as on 30th June 1980.

7. A proforma provide sheet of the company provides the following particulars :

	Amount per unit (Rs)
Element of cost	
Raw material	80
Direct labour	30
Overheads	60
Total	170
Profit	30
Selling price	200

The following further particulars are available :

Raw materials are in stock on average one month. Materials are in process, on average , half a month finished goods are in stock on average one month. Credit allowed by suppliers is one month. Credit allowed to debtors is two months. Log in payment of wages is $1\frac{1}{2}$ weeks. Log in payment of overhead expenses is one month.

One fourth of the output is sold against cash. Cash on hand and at bank is expected to be Rs. 25,000. You are required to prepare a statement showing the working capital needed to finance a level of activity of 1,04,000 units of production. You may assume that production is carried on evenly throughout the year, wages and overheads accure similarly and a time period of 4 weeks is equivalent to a month.

8. ABC ltd manufactures a single product for which demand exists for additional quantity. Present sale of Rs, 60,000 for month utilizes only 60% capacity of the plant. Sales manager assures that with a reduction of 10% in the price he would be in a position to increase the sale by about 25% to 30% The following data are available:

(a)	Selling price	Rs. 10 per unit
(b)	Variable cost	Rs. 3 per unit
(c)	Semi variable cost	Rs. 6,000 fixed plus Rs. 0.50 per unit
(d)	Fixed cost	Rs. 20,000 at present level estimated to be Rs. 24,000 at 80% output.

Determine the percentage increase in o/p which will be required to maintain present profit margin at the proposed selling price.

Modern traders are contemplating a reduction of price on their products by 10%. The following are the particulars.

	Rs.
Sales (20,000 units)	4,00,000
Fixed costs	80,000
Variable costs	15 per unit

It is also understood that by the use of machinery the variable costs will go down to Rs. 14 even though the fixed expenses will go up to Rs. 1,00,000.

How many units should be sold in order to maintain the same level of profit?

DE–3953 **43**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Solve the following Linear programming problem :

Maximize :

$z = 45x_1 + 80x_2$

Subject to the constraints :

$5x_1 + 20x_2 \leq 400$
$10x_1 + 15x_2 \leq 450$ (10)
$x_1, x_2 \geq 0$

(b) Solve the following Linear Programming problem

Minimize : $z = 3x_1 + 2x_2$

Subject to the constraints :

$x_1 - x_2 \leq 1$
$x_1 + x_2 \geq 3$. (10)
$x_1, x_2 \geq 0$

2. (a) Explain the Dual Simplex Algorithm in detail. (10)

(b) Discuss about the History and meaning of operational Research. (10)

3. Give a brief summary of various types of queuing models with suitable examples. (20)

4. Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponential, with mean 3 minutes.

(a) What is the probability that a person arriving at the booth will have to wait?

(b) What is the average length of the queue that form from time to time?

(c) The telephone department will install a second booth when convinced that an arrival would expect waiting for at least 3 minutes for phone. By how much should the flow of arrivals increase in order to justify a second booth? (20)

5. Customers arrive at a milk booth for the required service. Assume that inter-arrival and service times are constant and given by 1.8 and 4 times units, respectively.

Simulate the system by hand computations for 14 time units. What is the average waiting time per customer? What is the percentage idle time of the facility?

(Assume that the system starts at $t = 0$). (20)

6. A small marketing projects consists of the jobs in the table given below. With each job is listed its normal time and a minimum or crash time (in days). The cost (in Rs. per day) of crashing each job is also given.

Job $(i-j)$	Normal duration (days)	Minimum (Crash) duration (days)	Cost of crashing (Rs. per day)
1–2	9	6	20
1–3	8	5	25
1–4	15	10	30
2–4	5	3	10
3–4	10	6	15
4–5	2	1	40

(a) What is the normal project length and the minimum project length?

(b) Determine the minimum crashing costs of schedules ranging from normal length down to, and including, the minimum length schedule, that is, if L is the length of the normal schedule, find the costs of schedules which are $L, L-1, L-2$ and so on days long.

(c) Overhead costs total Rs. 60 per day. What is the optimal length schedule duration of each job for your solution? (20)

7. Explain the procedure for finding critical path in PERT networks with your own examples. (20)

8. A track owner finds from his past records that the maintenance costs per year, of a truck whose purchase price is Rs. 8,000 are as given below.

Year :	1	2	3	4	5	6	7	8
Maintenance Costs :	1000	1300	1700	2200	2900	3800	4800	6000
Resale price :	4000	2000	1200	600	500	400	400	400

Determine at which time it is profitable to replace the truck. (20)

DE–3954 **44**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Write an overview about the following :
   (a) Links and attributes. (5)
   (b) Generalization. (5)
   (c) Inheritance. (5)
   (d) Creation of object model. (5)

2. Discuss in detail the evolution of object model. (20)

3. Give detailed description about
   (a) relationship among classes (4)
   (b) identification of proper classes (4)
   (c) key abstraction mechanism. (12)

4. Describe the following with suitable examples :
   (a) Elaboration process (8)
   (b) Refactoring (4)
   (c) Meta model (6)
   (d) Inception. (2)

5. How to draw class diagram? Illustrate. (20)

6. Write detailed notes on UML programming. (20)

7. Explain the procedure for applying object model with an example. (20)

8. Draw state diagram and interaction diagram for banking process. (20)

DE–3955 **45**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Give an overview of internet programming. (10)

   (b) What are the design issues of WWW? Discuss. (10)

2. (a) Write a brief note on security and encryption. (10)

   (b) Briefly discuss on CGI. (10)

3. (a) Discuss briefly about developing intranet applications. (10)

   (b) Write a brief note on Java script. (10)

4. (a) Briefly discuss about Microsoft Internet explorer. (10)

   (b) Write a short note on HTML. (10)

5. (a) Give a brief note on VDO live technology. (10)

   (b) How do you pull web information? Discuss. (10)

6. (a) Discuss brief on creating an ActiveX control to activate a web page. (10)

(b) How do you create a custom integrated application with multiple protocols? Discuss. (10)

7. (a) Write a brief note on SGML. (10)

(b) Discuss briefly on images and hyperlinks. (10)

8. (a) How do you add graphics to web pages? Explain. (10)

(b) Write a brief note on animation with a suitable example. (10)

DE–3956 **51**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the components of Image processing system in detail with diagram. (10)

   (b) Describe the shape detection and recognition in detail. (10)

2. (a) Discuss about the perspective projection with examples. (10)

   (b) Write in detail about sampling and quantization with diagrams. (10)

3. (a) Explain the Restoration by Homomorphic filtering in detail. (10)

   (b) Describe the Low-pass filtering in detail. (10)

4. (a) Discuss about mean filter and ordered statistic filter. (10)

   (b) Write in detail about High-pass filtering. (10)

5. (a) Describe the lossy and lossless compression methods in detail. (10)

(b) Write in detail about the Huffman coding and Run-length coding. (10)

6. (a) Explain the processing of colour images and satellite images in detail. (10)

(b) Write in detail about Stereo Imaging Algorithms. (10)

7. (a) Discuss about the Local and Multi-level Thresholding. (10)

(b) Write in detail about Line Detection and Corner detection. (10)

8. (a) Explain the Region-based and Boundary-based description in detail. (10)

(b) Discuss about the Tree search and Graph matching. (10)

DE–3957 **52**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the size factors, quality and productive factors in detail.

   (b) Describe the Interactive enhancement model and Rapid Application development model in detail.

2. (a) Write in detail about process and project metrics.

   (b) Discuss about Linear sequential model and prototype model.

3. (a) Explain the Estimating software maintenance cost in detail.

   (b) Describe the programming Team structure in detail.

4. (a) Write in detail about scheduling plan.

   (b) Discuss about software Reviews.

5. (a) Explain the Risk Identification and projection in detail with diagrams.

(b) Describe the Risk Mitigation, monitoring and management in detail.

6. (a) Write in detail about software problem analysis.

(b) Discuss about the structured analysis.

7. (a) Discuss about Interface design guidelines.

(b) Explain the procedural design in detail.

8. (a) Describe the unit testing and Integration Testing in detail.

(b) Describe the white box testing and black box testing in detail with diagrams.

DE–3958 **53**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

VISUAL PROGRAMMING – II

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the Anatomy of C++ windows program. (10)

   (b) Describe the window messages in detail with examples. (10)

2. (a) Discuss about mainframe object and APP object. (10)

   (b) Write in detail about Document object and view object. (10)

3. (a) Explain the mouse Events in detail with examples. (10)

   (b) Describe the Adding shortcut keys in detail with examples. (10)

4. (a) Write in detail about the adding Accelerator keys with examples. (10)

(b) Discuss about adding menu items in code with examples. (10)

5. (a) Explain the dialog boxes in detail with examples. (10)

(b) Describe the message boxes in detail with example. (10)

6. (a) Write in detail about List boxes and combo boxes with examples. (10)

(b) Discuss about using VBX controls in code and graphics with examples. (10)

7. (a) Describe the sequential and Random access files in detail with examples. (10)

(b) Discuss about Adding scroll bars and splitter windows. (10)

8. (a) Explain the concept of Exception handling in detail with examples. (10)

(b) Describe the building ActiveX container Application in detail with examples. (10)

DE–3959 **54**

DISTANCE EDUCATION

M.C.A (S)/M.C.A (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Write a detailed note on Intermediate code generation. (10)

   (b) Explain the use of Table management and Error Handling. (10)

2. (a) Explain with suitable examples the construction of Parse Tree. (15)

   (b) Define context free Grammar. Give an example. (5)

3. Describe Syntax directed translation scheme and its implementation with relevant examples. (20)

4. Discuss in detail about lexical phase errors and Syntactic phase errors. (20)

5. Explain in detail about

   (a) Peephole optimization (10)

   (b) Basic blocks and Flow graphs. (10)

6. (a) Write a detailed note on role of lexical analyzer. (10)

(b) Give a detailed note on Three address code, with example. (10)

7. (a) Explain briefly about Semantic Errors. (10)

(b) Convert the assignment statement X=(A+C)–(C–D)–(C–D) into three address statement. (10)

8. Explain in detail about Top down Parsing with example. (20)

DE–3960 **55**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Discuss about data mining versus query tools. (10)

   (b) Discuss about self learning computer systems. (10)

2. (a) Write in detail about data selection and data cleaning. (10)

   (b) Explain the visualization techniques in detail. (10)

3. (a) Describe the decision trees in detail with examples. (10)

   (b) Discuss about the association rules. (10)

4. (a) Write in detail about Data warehouse architecture with diagram. (10)

   (b) Discuss about Aggregation and Data marting. (10)

5. (a) Explain the database scheme in detail. (10)

   (b) Describe the systems and data warehouse process managers in detail. (10)

6. (a) Write in detail about service level agreement. (10)

(b) Discuss about the backup and recovery in datawarehouse. (10)

7. (a) Discuss about the security concept in data warehouse. (10)

(b) Describe the hardware and operational design of data warehouses in detail. (10)

8. (a) Explain the data warehouse features in detail. (10)

(b) Write in detail about testing the data warehouses. (10)

DE–3961 **61**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Discuss the important activities covered by the software project management.

2. Explain briefly about Network Planning model with a neat sketch.

3. Discuss in detail about the process of Risk Analysis and management.

4. What is meant by Prototyping? Explain the various models of prototyping with suitable illustrations.

5. Explain briefly about the important objectives of prince project management.

6. Discuss briefly about the resource allocation with neat sketch.

7. Why Lines of Code (LOC) metric not considered as an estimate for project size? Justify your answer.

8. Write a brief note on various steps involved in Project Planning.

DE–3962 **62**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the commercial tools in multimedia in detail.

   (b) Describe the need of multimedia and problems of multimedia in detail.

2. (a) Discuss about the multimedia platforms.

   (b) Write in detail about system components in multimedia.

3. (a) Explain the concept of Animation in detail with diagrams.

   (b) Discuss about text, image and graphics in multimedia.

4. (a) Describe the video compression techniques in detail.

   (b) Discuss about JPEG and DVI.

5. (a) Write in detail about redundancy and visibility.

(b) Explain the MPEG (Motion Picture Expert Group) in detail.

6. (a) Explain the object and classes in object oriented multimedia in detail.

(b) Describe the media classes in multimedia frameworks in detail.

7. (a) Write in detail about Transform classes in multimedia frameworks.

(b) Discuss about Format Classes in multimedia frameworks.

8. (a) Explain the multimedia on networks in detail.

(b) Describe the Architecture and operations in multimedia environments in detail.

DE–3963 **63**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the biological neural networks in detail with diagram. (10)

   (b) Describe the construction of neural networks in detail with diagram. (10)

2. (a) Write in detail about the applications of artificial neural networks. (10)

   (b) Discuss about neuron models with diagram. (10)

3. (a) Explain the supervised learning concepts in detail. (10)

   (b) Describe the unsupervised learning concepts in detail. (10)

4. (a) Write in detail about back propagation networks with diagram. (10)

   (b) Discuss about the counter propagation networks with diagram. (10)

5. (a) Discuss about the concept of perceptrons. (10)

(b) Explain the applications of back propagation networks in detail with diagrams. (10)

6. (a) Explain the hopfield network in detail with diagram. (10)

(b) Describe the associative memory in detail with diagram. (10)

7. (a) Discuss about the applications of associative memory. (10)

(b) Explain the bidirectional associative memory in detail. (10)

8. (a) Explain the concept of adaptive resonance theory in detail. (10)

(b) Describe the neocognitron concepts in detail with diagrams. (10)

DE–3964 **64**

DISTANCE EDUCATION

M.C.A. (S) / M.C.A (S)(Lateral) DEGREE
EXAMINATION, MAY 2012

Sixth Semester

C SHARP (C#)

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about the evolution of C#. (10)

   (b) Write in detail about the components of .NET. (10)

2. (a) Explain the common language runtime and class library in detail. (10)

   (b) Describe the automatic resource management in detail. (10)

3. (a) Discuss about the virtual object system in C# (10)

   (b) Write in detail about the features of C# (10)

4. (a) Explain the concept of struct in C# in detail with example program. (10)

   (b) Describe the boxing and unboxing in C# in detail with example programs. (10)

5. (a) Write in detail about constructors in C# with example programs. (10)

(b) Discuss about the destructors in C# with example programs. (10)

6. (a) Explain the concept of Exception handling in C# in details with example programs. (10)

(b) Discuss about the concept of working with Namespaces in C# with example programs. (10)

7. (a) Describe the control statements in C# in detail with example programs. (10)

(b) Discuss about the concept of Iteration and selection in C# with example programs. (10)

8. (a) Write in detail about configuring and deployment in C#. (10)

(b) Discuss about platform invocation services in C#. (10)

DE–4391 **36**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION,

MAY 2012.

Third Semester

LAB V — RDBMS

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Student has to answer ONE question by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. An airline reservation database contains the reservation and personal table.

Reservation Table contains following fields :

Flight number, class, passenger name, seat number.

Personal Table contains following fields :

Passenger Name, Sex, Age, Martial status, Nationality

Write a program in ORACLE to prepare the following reports :

(a) List the passengers names with seat number according to class and flight numbers.

(b) Total number of married female passengers in a particular flight.

---------- Cut here ----------

2. Write a PLSQL program to prepare a mark statement of a student using student table and mark table. Assume your own data.

3. A salary statement contains Name, Basic pay, Allowances, Total Deduction (includes IT), Grosspay, Netpay.

Dearness Allowance = 58% of Basic pay

Gross pay = Basic pay + D.A

Deduction = P.F. + LOAN AMOUNT

Income tax is calculated on the basis of Annual Income with the following conditions.

Annual Salary		Income tax
< = 1 lakh		Nil
> 1 lakh but < = 1.5 lakh	–	10%
> 1.5 lakh but < = 3 lakh	–	20%
> 3 lakh and above	–	30%

Total deduction = Deduction + Income Tax

Write a program in SQL to prepare salary report for 25 employees.

---------- Cut here ----------

4. Create a table for savings bank account to customer with suitable fields. Write PL/SQL program to perform money deposit and withdrawal operations.

5. Create a salesman table with the following fields :

SMAN

SM Number

Name

Sale amount

Product code

Date of sale

Write PL/SQL program to find the following

(a) Total sales salesman wise

(b) Total sales product wise

---------- Cut here ----------

6. Create three tables in oracle with the following fields :

Cricket	Football	Hockey
Player No:	Player No:	Player No:
Name	Name	Name
Age	Age	Age
Number of Matches	Number of Matches	Number of Matches
Runs	Goals	Goals

(a) Write query to display the players playing all the three games.

(b) Write query to display the cross product of all the tables.

7. A hospital maintains blood donors record information in a file. The file contains the following fields :

Donor number, Donor name, Donor age, Donor Address, pin code, place of birth, blood group (A, B, AB and O)

Write a program using oracle to print out the number, name and address of the donors for the following categories.

(a) Male donors with 'A' blood group in the age between 18 and 25

(b) Female donors with 'A' blood group in the age between 18 and 25.

---------- Cut here ----------

8. Write a program in SQL for Hospital Billing system with the following fields : Patient no, patient name, age, attending doctor, Patient-type (IN/OUT), consulting charge, Blood test charge, X-Ray charges, other test charges and total fee.

Prepare Reports with the following conditions :

(a) Patient undergone blood test

(b) The patient, who taken X-rays

The common fields to be included in the above mentioned reports : patient no., patient name, age, charges and total charges.

9. Write a PL/SQL program to prepare a monthly attendance statement of a student using student table and attendance table. Assume your own data.

---------- Cut here ----------

10. An examinations has been conducted to a class 55 students and the following are given as inputs :

Three test marks, Register No. and candidate name

Write an SQL program do the following :

Assign a letter grade to each student based on the average score and list Register Number, Average store, grade. The minimum pass for each subject is 50. The Grading system :

Average	Score	Grade
90–100	–	A
75–89	–	B
60–74	–	C
50–59	–	D
0–49	–	F – Fail

DE–4392 **37**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Third Semester

LAB VI — ALGORITHMS AND SHELL PROGRAMMING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

One question is to be given by the examiner to each candidate by lot system.

Break up of marks:

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Write a program using C/C++ to find the Fibonachi series using recursion.

   (b) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

      (i) File writable

      (ii) Both readable and writable.

2. (a) Write a program using C/C++ to find the Ackerman's function using recursion.

(b) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

(i) Enter the five names in a file

(ii) Sort the names in ascending order in existing file.

---------- Cut here ----------

3. (a) Write a program using C/C++ to solve the towers of Hanoi problem.

(b) Write a shell program which accepts the name of a file from the standard input and then perform the following tests on it :

(i) Sort the names in descending order in the existing file

(ii) Search a given whole word in an existing file

(iii) Exit.

---------- Cut here ----------

4. (a) Write a program in C/C++ for straight selection sort. Test the program with sample data.

(b) Write and execute shell program to send mail to groups of users by extracting their ids form/etc/ group mail.

---------- Cut here ----------

5. (a) Write a program in C/C++ for straight insertion sorting. Test the program with sample data.

(b) Write and execute shell program to generate leave status report of an organization.

6. (a) Write a program in C/C++ for straight exchange sorting. Test the program with sample data.

(b) Write and execute a shell program to print monthly payships for employees in an organization. Assume your own data and generate accordingly.

---------- Cut here ----------

7. (a) Write a program using C/C++ for heap sort. Test the program with sample data.

(b) Write a program for four-function calculator by using yacc command.

---------- Cut here ----------

8. (a) Write a program using C/C++ for the knight tour problem using backtracking method.

(b) Write a shell program to sum up the series $\frac{1}{1!}+\frac{1}{2!}+.....+\frac{1}{10!}$.

---------- Cut here ----------

9. (a) Write a program using C/C++ to perform leap sorting. Test the program with sample data.

(b) A hospital maintains patient details. Write shell program a list the following:

(i) Patients in a blood groupwise

(ii) Patients in age between 20 to 30

(iii) Exit.

10. (a) Write a program using C/C++ for quick sort (Divide and conquer). Test the program with sample data.

(b) Write a shell program to prepare electric bill for domestic consumers.

For first 100 units – Rs. 0.75/unit

For next 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit.

Prepare the electricity bill in the following format :

Customer No. ———

Customer Name ———

Previous reading ———

Current reading ———

Units consumed ———

Charge ———

Assume your own data and generate accordingly.

———————

DE–4393 **46**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any ONE question chosen by the examiner through lot system.

Each subdivision carries 50 marks.

Breakup of marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a Java script to create a book issue form for the library.

   (b) Develop a single web page form to display the details of the student.

---------- Cut here ----------

2. (a) Write a Java script to create a window by using the confirm message.

   (b) Write a VB script to do the following :

      (i) Check the given password

      (ii) Change the existing password.

3. (a) Write a Java script to create a hostel room request form to the hostel office.

(b) Write a VB script to do the following :

(i) Display the current date.

(ii) Find the difference between two dates.

(iii) Find the age of a person by providing date of birth. For all use the date in dd/mm/yy format.

---------- Cut here ----------

4. (a) Write a VB script to create an electronic watch having the provision to display hour, minutes and seconds.

(b) Write a Java script to create a Railway Reservation form for ticket booking.

---------- Cut here ----------

5. (a) Write a Java script to create a Bus Reservation form for ticket booking.

(b) Write a menu driven Java program to do database functions using database connectivity facility. The functions include display, edit, insert and delete a records from a table.

---------- Cut here ----------

6. (a) Write a Java script to create a Flight Reservation form for ticket booking.

(b) Write a Java program using Applet to display the dialogue menu in Applet.

---------- Cut here ----------

7. (a) Write a Java swing program to create buttons with (i) Tool tip text (ii) Image.

(b) Develop a single page advertisement about your college and the course availability details in the college.

8. (a) Write a Java program using Applet to display the different colors and fonts.

(b) Develop a web page that contains the details about the information of your department faculty members.

---------- Cut here ----------

9. (a) Write a Java program to create a color as the background.

(b) Using atleast 20 HTML tags, create a screen with a string "WEB DESIGN".

---------- Cut here ----------

10. (a) Write a swing program to create the Tabbed panels.

(b) Develop a web page for a Bonafide certificate used for the identity of the students.

DE–4394 **47**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fourth Semester

LAB VIII — NETWORK LAB

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any ONE question chosen by the examiner through lot system.

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Write a Java program to find the square root of a number using RMI client, server technique.

   (b) Write a Java program to add two numbers using RMI client, server technique.

---------- Cut here ----------

2. (a) Write a Java program to find factorial of given number using RMI client, server technique.

   (b) Write a Java program to find the cube root of a number using RMI client, server technique.

3. (a) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

(b) Write a Java program to find the square of given number using RMI client, server technique.

---------- Cut here ----------

4. (a) Write a Java program to setup a chat server and chat client.

(b) Write a Java program to setup an echo server and echo client.

---------- Cut here ----------

5. (a) Write a Java program to display the users and their IP addresses logged in the network.

(b) Write a Java program to write a file from the remote system.

---------- Cut here ----------

6. (a) Write a Java program to send message from one system to another system.

(b) Write a Java program to ENCRYPT given word.

---------- Cut here ----------

7. (a) Write a Java program to setup time server and time client.

(b) Write a Java program to send messages from one system to all the systems in the LAN network.

---------- Cut here ----------

8. (a) Consider a LAN network consists of six systems named as L1, L2, L3, L4, L5 and L6. Write a Java program to send message from L1 system to L3 and L5.

(b) Write a Java program to read a file from the remote system.

9. (a) Write a Java program to find the cube of a number using RMI client, server technique.

(b) Write a Java program to decrypt the given encrypted word.

---------- Cut here ----------

10. (a) Write a Java program to find the inverse of the given number using RMI client, server technique.

(b) Write a Java program to find the fourth power of the given number using RMI client, server technique.

DE–4395 **56**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

LAB IX — VISUAL C++

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system.

Each question carries 50 marks.

Break-Up of Marks	:	
Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write visual C++ win 32 application program using MFC that allows the user to draw pictures with the help of mouse as a free hand drawing tool.

   (b) Write visual C++ win 32 application program using MFC that fills the background of the client area with a bitmap.

2. (a) Write visual C++ win 32 application program using MFC that displays the status of ALT, CTRL, SHIFT, NUMLOCK and SCROLL lock keys.

(b) Write visual C++ win 32 application program using MFC that creates to push buttons OK and CANCEL on the client area. Buttons should respond to user click over them and display appropriate message.

---------- Cut here ----------

3. (a) Write visual C++ win32 application program using MFC that displays a menu. Choose the menu items using keyboard accelerator keys and display appropriate messages for the selected command in message box.

(b) Write visual C++ win 32 application program using MFC that display "Alagappa university".

---------- Cut here ----------

4. (a) Write visual C++ program win 32 application using MFC that creates a new font.

(b) Write visual C++ program win 32 application using MFC that bills the back ground of the client area with bit map.

DE–4396 **57**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Fifth Semester

LAB X — COMPILER DESIGN

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break - up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Write a program in C language to recognize keywords in the given sources program.

   (b) Write a program in C language to recognize identifier in the given sources code.

---------- Cut here ----------

2. (a) Write a program in C language to recognize constant in the given source code.

   (b) Write a C program to accept a string and check whether it satisfies the following production rule (By using shift reduce parsing)

   $E \rightarrow E + E / E - E / E*E / E / E / id / Const / (E)$.

3. Write a C program to check the production rule whether it is an operator grammar or not

(a) $S \rightarrow aBcd$

$S \rightarrow aBdc$

(b) $E \rightarrow E+E/E*E/(E)/id$

(c) $E \rightarrow E\ OP\ EE \rightarrow (E)$

$E \rightarrow id$

$OP \rightarrow + /- /*/ /.$

---------- Cut here ----------

4. Construct a C program to accept a string and check whether it satisfied the following production rule (By using operator precedence grammar)

$S \rightarrow AbB$

$A \rightarrow aBb$

$B \rightarrow a/Bba.$

---------- Cut here ----------

5. Write a C program to accept a string and check whether it belongs to ambiguous grammar or not by using the following production rule.

$E \rightarrow E+E/E-E/E*E/E/E/id/Const/(E).$

---------- Cut here ----------

6. Write a C program to construct the left-most derivation and right-most derivation for the following grammar

$S \rightarrow a/^/(T)$

$T \rightarrow T, S/S$

(a) (a, (a, a))

(b) To find the LEADING and TRAILING sets.

7. Write a C program to construct predictive parsing table for the following production rule

$S \rightarrow i(tSS1/a$

$Si \rightarrow eS/E$ (E is empty)

$C \rightarrow b$

---------- Cut here ----------

8. Write a C program to construct a SLR parsing table for the following grammar

$E \rightarrow E+T/T$

$T \rightarrow T*F/F$

$F \rightarrow (E)$ /id.

---------- Cut here ----------

9. Write a C program to draw the transition diagram for DFA by accepting the following regular expression

(a) (a/b) *abb

(b) (1/0)1+.10 (1/10) *.

---------- Cut here ----------

10. Write a C program to construct a canonical LR parsing table for the following grammar

$S \rightarrow CC$

$C \rightarrow cC/d.$

DE-4397 **65**

DISTANCE EDUCATION

M.C.A (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2012.

Sixth Semester

LAB XI — MULTIMEDIA APPLICATIONS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot System.

Each sub division carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Prepare a multimedia presentation that shows the features of your department staff members.

   (b) Design a friends day greeting page including various graphics feature in flash.

2. (a) Design a placement brochure advertisement for your class placement using flash.

(b) Prepare a multimedia presentation that shows the features of your department association function.

---------- Cut here ----------

3. (a) Design a pongal festival greeting page including various graphic feature in flash.

(b) Prepare a multimedia presentation that shows the features of your college courses offered.

---------- Cut here ----------

4. (a) Design a Deepavali festival greeting page including various graphic feature in flash.

(b) Prepare a multimedia presentation that shows the features of your college library.

---------- Cut here ----------

5. (a) Write a C++ program for setting up a pattern for multi-grey level images or bilevel display using limbs algorithm with the following 2×2 matrix.

0	2
3	1

Also print the dot pattern produced on a 16 grey-level system for a pixel.

(b) Design an animated cartoon picture.

---------- Cut here ----------

6. (a) Do the image animation using flash.

(b) Write a C++ program for 2-dimensional image animation using computer graphics.

7. (a) Write a C++ program that smoothens an image in the transform domain.

(b) Write a C++ program to enhance an image using median filter.

---------- Cut here ----------

8. (a) Write a C++ program to perform image subtraction operation.

(b) Write a C++ program to compress an image using contour coding technique.

---------- Cut here ----------

9. (a) Write a C++ program for grey level transform techniques (i) Thresholding (ii) Bunching.

(b) Write a C++ program for convolution using given template.

Image					Template	
1	1	3	3	4		
1	1	4	4	3	1	0
2	1	3	3	3	0	1
1	1	1	4	4		

---------- Cut here ----------

10. (a) Write a C++ program for creating an image from the given image using automatic selection of grey level for splitting technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

(b) Write a C++ program to find the edges of the following image using sobel edge detection procedure

0	0	0	0	0	2	2	0	3	3
0	0	1	0	0	0	2	4	2	2
0	2	0	0	2	4	3	2	3	3
0	1	3	3	4	4	3	3	3	3
0	1	0	4	3	3	2	4	3	3
1	0	0	1	2	3	4	2	3	4